

CHARLES XU

charlsxu@mit.edu | github.com/HungryAmoeba

EDUCATION

Massachusetts Institute of Technology

Sept. 2024 – Present

PhD in Computer Science, expected September 2029

Advisor: Jörn Dunkel (Professor of Mathematics)

Master of Science (SM), earned en route to PhD

May 2026

Thesis: Algebra Discovery via Free Algebra Relations

Yale University

May 2023

BS Applied Mathematics; BS Chemistry

RESEARCH INTERESTS

I study how to infer interaction laws and governing dynamics from observations of biological and physical systems. I also develop mathematically interpretable representations of these systems using geometric algebra and geometric machine learning.

SELECTED RESEARCH EXPERIENCE

Graduate Researcher, MIT

Sept. 2024 – Present

Advisor: Jörn Dunkel

- Investigating methods to infer interaction laws and governing dynamics of interacting systems, including animal behavior.
- Developing interpretable representations of biological and physical systems using geometric algebra.
- SM thesis: Developed a method to learn operator algebras from experimental data by constructing the ideal of relations from the null space of an experimental data matrix.

Post-baccalaureate Research Assistant, Yale

Aug. 2023 – Aug. 2024

Krishnaswamy Lab; Advisor: Smita Krishnaswamy

- Developed geometric deep learning methods for graph signals and high-dimensional biological data, including hypergraph wavelets for spatial transcriptomics.
- Contributed to the mathematical development of DYMAG and implemented chaotic dynamics for graph message passing.

Sweeney Lab, Yale

Sept. 2021 – May 2023

Advisor: Alison Sweeney

- Used Monte Carlo simulations in C to study semiflexible polymers on cylindrical surfaces under confinement.
- Developed image-processing algorithms to characterize polymer network topology in cryoEM images.

TECHNICAL SKILLS

Programming: Python, C, MATLAB, R.

Tools: PyTorch, JAX, PyTorch Geometric, TensorFlow, Git, $\text{\LaTeX}$.

PUBLICATIONS

DYMAG: Rethinking Message Passing Using Dynamical-systems-based Waveforms.

Bhaskar, D.* , Sun, X.* , Zhang, Y.* , **Xu, C.*** , Afrasiyabi, A., Viswanath, S., Fasina, O., Wolf, G., Perlmutter, M., and Krishnaswamy, S.

Topology, Algebra, and Geometry in Data Science (TAG-DS), 2025. **Spotlight.**

Proceedings of Machine Learning Research 321:236–268, 2026.

Hyperedge Representations with Hypergraph Wavelets: Applications to Spatial Transcriptomics.

Sun, X.* , **Xu, C.*** , Rocha, J. F., Liu, C., Hollander-Bodie, B., Goldman, L., DiStasio, M., Perlmutter, M., and Krishnaswamy, S.

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025.

Principal Curvatures Estimation with Applications to Single Cell Data.

Zhang, Y.* , Mezrag, L.* , Sun, X., **Xu, C.**, Macdonald, K., Bhaskar, D., Krishnaswamy, S., Wolf, G., and Rieck, B.

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025.

BLIS-Net: Classifying and Analyzing Signals on Graphs.

Xu, C., Goldman, L., Guo, V., Hollander-Bodie, B., Trank-Greene, M., Adelstein, I., De Brouwer, E., Ying, R., Krishnaswamy, S., and Perlmutter, M.

International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.

Proceedings of Machine Learning Research 238:4537–4545.

**Equal contribution.*

TEACHING AND MENTORING

Tutor, Research Science Institute (RSI)

2026

Mentored computational biology students on the research process, scientific writing, and presentations.

Undergraduate Mentor, MIT UROP

Fall 2024

Provided background instruction on graph neural networks for single-cell analysis through weekly meetings and instructional materials.

Teaching Assistant, Yale

Advanced Linear Algebra with Applications

Spring 2024

Probability and Statistics

Fall 2022

Deep Learning Theory and Applications

Spring 2022

SELECTED TALKS

“Why Do Plants Look Like That?”

April 2026

Simple Persons Applied Math Seminar (SPAMS), MIT

“Hyperedge Representations with Deep Hypergraph Wavelets: Applications to Spatial Transcriptomics”

June 2024

Graph Signal Processing Workshop, Delft, Netherlands

SERVICE AND LEADERSHIP

Reviewer: NeurIPS (2024); Sampling Theory and Applications (2023).

Social Chair, MIT EECS Graduate Student Association

2026 – Present

IT Chair, Edgerton Residence, MIT

April 2026 – Present

Mentor, MIT Graduate Application Assistance Program

2024 – Present